Supplement to TrajCert: Sharp risk identification and anytime-valid certification with informative outcome-resolution paths

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This supplement supplies analytical details, conservative numerical arguments, the complete final-cell power results and reproducibility information. Notation agrees with the article: A, G, c are resolved harmful, resolved beneficial and terminal unresolved masses, u is terminal harmful mass, and $r = A + G$.

1 Information decomposition and all boundary cases

For fixed observable masses, every completion has the joint table

$$\begin{array}{c|cccc} & 1 & \cdots & K & \infty \\ \hline L=1 & a_1 & \cdots & a_K & u \\ L=0 & b_1 & \cdots & b_K & c-u \end{array}. \tag{S1}$$

Its entries are nonnegative and sum to one exactly when $0 \le u \le c$. The row harmful marginal is $A + u$. Conditioning on columns gives

$$H(L \mid J_\Pi) = \sum_{k:a_k+b_k>0} (a_k + b_k) h\left(\frac{a_k}{a_k + b_k}\right) + q(u, c-u). \tag{S2}$$

This proves the information profile in the article without a selection-model assumption. When $r, c > 0$, put $p_R = A/r$, $p_U = u/c$ and $\theta = rp_R + cp_U$. Then

$$\mathcal{S}_\Pi(u) - \tau_\Pi = h(\theta) - rh(p_R) - ch(p_U) \tag{S3}$$

$$= rD_{\mathrm{KL}}\{\mathrm{Bern}(p_R) \| \mathrm{Bern}(\theta)\} + cD_{\mathrm{KL}}\{\mathrm{Bern}(p_U) \| \mathrm{Bern}(\theta)\} \ge 0. \tag{S4}$$

The equality is interpreted by continuity if a marginal is zero. For $0 < r, c < 1$, equality in the divergence inequality holds precisely at $p_R = p_U$, proving $u = Ac/r$. Thus τ_Π is the minimal possible full information and is determined by the resolved timing distribution. Equation (S4) also displays the model as an information budget for terminal selection after accounting for observed timing.

For interior hidden masses, $h'(x) = \log\{(1-x)/x\}$ and $h''(x) = -1/\{x(1-x)\}$. Hence

$$\mathcal{S}''(u) = \frac{1}{u} + \frac{1}{c-u} - \frac{1}{A+u} - \frac{1}{G+c-u} = \frac{A}{u(A+u)} + \frac{G}{(c-u)(G+c-u)}. \tag{S5}$$

If $r > 0$, at least one numerator is positive, so the profile is strictly convex whenever $c > 0$. If $A = 0 < G$, the derivative is $\log\{(G+c-u)/(c-u)\} > 0$ and the minimum is $u = 0$. If $G = 0 < A$, the derivative is $\log\{u/(A+u)\} < 0$ and the minimum is $u = c$. If $c = 0$, the only completion is $u = 0$, with information $h(A) - T_\Pi = \tau_\Pi$. If $r = 0$, $c = 1$, the profile is $h(u) - h(u) = 0$ throughout $[0, 1]$. These cases also show why no minimum-information risk is defined from resolved cases when $r = 0$.

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For $r > 0, c > 0$, strict convexity and continuity imply that the sublevel set is empty below its minimum, is the unique minimizer at equality, and is a nonempty compact interval above it. It can touch either boundary; an interior root need not exist on both branches. Since every feasible completion is a valid table and no other completion is possible, its risk interval is sharp. If only coarse observations are supplied, any completion can be lifted to a hypothetical finer representation with no additional within-coarse-band outcome dependence, preserving mutual information. Known fine margins restrict such a lift, which is why the article states sharpness with respect to the supplied partition.

2 Refinement, strictness and decision frontiers

Let each finite coarse band B collect fine bands with masses $m_k = a_k + b_k$ and harmful proportions $r_k = a_k/m_k$. For $m_B = \sum_{k\in B} m_k > 0$, its harmful proportion is $\bar{r}_B = \sum_{k\in B} m_k r_k / m_B$. Its entropy gain is

$$m_B h(\bar{r}_B) - \sum_{k\in B} m_k h(r_k) \geq 0. \tag{S6}$$

Strict concavity makes the inequality strict exactly when positive-mass proportions within B differ. Sum over coarse bands. The unresolved column is unchanged, so this sum is $T_c - T_f$, independent of its hidden label split. The chain rule for the deterministic map $J_c = g(J_f)$ gives

$$I(L; J_f) = I(L; J_c) + I(L; J_f \mid J_c), \tag{S7}$$

and identifies the sum with the conditional information. Comparing profiles gives nesting at every common budget, including empty-set cases.

For the sufficient strictness condition in the article, fine compatibility gives $\rho \geq \tau_f > \tau_c$. Therefore the coarse right endpoint is strictly to the right of its minimizer. If it is below c, its coarse profile value is ρ and its fine profile value is $\rho + \Delta$. Fine feasibility excludes that point. Compactness prevents the fine upper endpoint from approaching it without attaining it, proving strict inequality. If both right endpoints are c, the upper risks coincide despite $\Delta > 0$. Thus informative timing is insufficient for strict improvement when both bounds are saturated. If $c = 0$, there is no hidden risk to tighten.

For the frontier, first note that every compatible set contains $\theta^\dagger = A/r$. Hence $\beta < \theta^\dagger$ is intrinsically uncertifiable even if $A \leq \beta$; $\beta < A$ is the stronger direct obstruction. If $\beta \geq A + c$, every completion is safe. Otherwise, $\theta^\dagger \leq \beta < A + c$ puts $u_\beta = \beta - A$ on the right branch, where the profile is strictly increasing beyond its minimum. If $\rho \leq \mathcal{S}(u_\beta)$, its sublevel-set right endpoint is at most u_β. If $\rho > \mathcal{S}(u_\beta)$, continuity gives a feasible point to the right of u_β (as $u_\beta < c$), so certification fails. Equality therefore belongs to the certifying side. Subtracting the two profiles at u_β gives the exact frontier shift. In the case $\beta = \theta^\dagger < A + c$, the only certifying budget is $\rho = \tau$.

3 Proof of the simultaneous and projected guarantee

For a true category probability $p_j \in (0, 1)$, write $X_{j,i} = \mathbf{1}\{Y_i = j\}$. For any fixed $z \in [0, 1]$, the process

$$Z_n(z; p_j) = \prod_{i=1}^{n} \left(\frac{z}{p_j}\right)^{X_{j,i}} \left(\frac{1-z}{1-p_j}\right)^{1-X_{j,i}} \tag{S8}$$

is a nonnegative martingale, since its next multiplicative factor has conditional expectation $p_j(z/p_j) + (1 - p_j)\{(1-z)/(1-p_j)\} = 1$. Integrating against a Beta$(1/2, 1/2)$ law preserves this property by conditional Tonelli, with initial value one, and gives the article's $M_{j,n}(p_j)$. At $p_j = 0$, all $X_{j,i} = 0$ almost surely and the limiting mixture is nonincreasing with initial value one; $p_j = 1$ is analogous. The maximal inequality therefore gives

$$\mathbb{P}\{\exists n : M_{j,n}(p_j) \geq d/\delta\} \leq \delta/d. \tag{S9}$$

A union bound over the d categories proves simultaneous containment in every raw interval with probability at least $1 - \delta$. Taking running intersections and intersecting with the simplex preserve this same event. Categories within an event need not be independent. No union bound over monitoring times is used.

On simultaneous containment, the true summary belongs to the exact observable-law region at every n. The actual terminal harmful mass supplies a feasible completion because its finest-path information is at most ρ, and coarse information cannot be larger. Therefore θ is contained in the union of partition-specific risk sets at every n. Its supremum is bounded by the aggregate relaxation and by any conservative numerical enclosure. This proves the article's time-uniform risk guarantee. Restricting reports to checkpoints or adding evidence gates can only remove opportunities to certify, so neither requires extra error spending. If the sensitivity assumption fails, this feasibility step fails; compatibility of some other completion does not repair it.

The event also implies simultaneous coverage of every member of the partition-specific identified set: on it, the entire true observable-law set is part of each projection. Thus the guarantee is stronger than pointwise containment of a selected terminal completion, while the implemented relaxation may be wider than necessary. Across multiple partitions or multiple streams the same reasoning needs a simultaneous observable region or an explicit union allocation; the single-partition theorem does not supply such an allocation automatically.

4 Envelope and branch-and-bound derivation

For any selected set of categories B, simplex membership implies

$$\max\{\sum_{j\in B}\ell_j, 1-\sum_{j\notin B}v_j, 0\} \le \sum_{j\in B}p_j \le \min\{\sum_{j\in B}v_j, 1-\sum_{j\notin B}\ell_j, 1\}. \tag{S10}$$

These inequalities give the aggregate mass bounds. For $a, b > 0$, $\partial q/\partial a = \log\{(a+b)/a\} \ge 0$ and $\partial q/\partial b = \log\{(a+b)/b\} \ge 0$. Continuity extends monotonicity to boundaries. Thus evaluating q at simultaneous lower or upper coordinates gives valid lower or upper entropy bounds. The raw upper sum is valid but can be loose; intersecting it with the resolved-mass entropy ceiling and $\log 2$ tightens it. A ceiling is not needed to turn the raw sum into an upper bound. This monotonicity argument, rather than attainability of simultaneous upper coordinates, justifies the envelope.

Every categorical summary maps into $\mathcal{E}_n$. Since $\mathcal{S} = h(A+u) - T - q(u, c-u)$ decreases in T, replacing the actual T by T_U can only enlarge its feasible set. For a search box $B = [A_-, A_+] \times [G_-, G_+] \times [u_-, u_+]$, define

$$c_- = \max(c_L, 1 - A_+ - G_+, 0), \qquad c_+ = \min(c_U, 1 - A_- - G_-, 1), \tag{S11}$$

$$\tilde{u}_+ = \min(u_+, c_+), \qquad z_- = \max(0, c_- - \tilde{u}_+), \quad z_+ = \max(0, c_+ - u_-). \tag{S12}$$

Impossible mass constraints discard the box. Otherwise every feasible point has $u \in [u_-, \tilde{u}_+]$, terminal beneficial mass in $[z_-, z_+]$, and $\theta \in [A_- + u_-, \min(1, A_+ + \tilde{u}_+)]$. A lower enclosure is

$$\underline{S}_B = \max\left\{0, \inf_{x\in[A_-+u_-, \min(1, A_++\tilde{u}_+)]} h(x) - T_U - \sup_{u\in[u_-,\tilde{u}_+],\, z\in[z_-,z_+]} q(u,z) - \eta_B\right\}, \tag{S13}$$

Here $\eta_B = 0$ for the revised implementation: every displayed arithmetic bound is evaluated in the enclosing direction rather than corrected by an empirical allowance. The archived implementation instead used $32(K+1)\,\mathrm{ulp}(1)$ for specified floating operations. The binary-entropy infimum is at an interval endpoint. The mass entropy q is increasing in each argument and is continuous at zero; lower corners therefore give lower bounds. Its upper bound is the smaller of a direct Arb interval upper bound, when both arguments are bounded away from zero, and $(u_+ + z_+)\log 2$. Taking the maximum with zero need not bound a negative relaxed expression from below. It nevertheless preserves every feasible relaxed point when $\rho \ge 0$, since replacing a negative lower bound by zero cannot make it exceed ρ. Thus a box with $\underline{S}_B > \rho$ contains neither a feasible point of V_n nor an original compatible completion.

The following argument covers the revised functions in `confidence.py`, `envelope.py`, `projection.py` and `certification.py`. Its arithmetic assumptions are Python's exact integer/rational operations, correctly rounded binary64 conversion and division, adjacent-float stepping, and the enclosure contract of Arb (Johansson, 2017, in the article reference list). Supplied binary64 endpoints and budgets denote their exact dyadic values. Endpoints are finite and lie in $[0, 1]$, tolerances are finite and positive, and each process

executes the numerical calls serially without external mutation of the Arb context. The repository uses separate processes for parallel streams. The argument is a source-level numerical proof under these runtime contracts, not a machine-checked proof of the language, libraries or hardware.

Directed arithmetic and confidence input. For an exact rational v, the code first converts to a nearest binary64 value f, then compares f with v exactly. A lower conversion moves to the preceding float only when $f > v$; the upper conversion is symmetric. Exact rational sums followed by these conversions enclose all mass sums, complements, objectives and reported gaps. An Arb lower or upper endpoint is first extracted as an exact integer times a power of two and then converted in the same direction. Nonfinite Arb results give infinite bounds or a numerical failure, never a favorable sign. For each category, the beta-mixture log likelihood minus $\log(d/\delta)$ is evaluated as an Arb ball, including the beta-function term and threshold. This function is convex in the probability. Inversion maintains an outer boundary point and a certified nonpositive inner point on each side; a positive interval sign moves the outer point, and a nonpositive interval sign moves the inner point. An unresolved sign leaves the outer bracket in place. If no inner sign is certified, the entire corresponding side $[0, \widehat{p}]$ or $[\widehat{p}, 1]$ is retained. Hence the returned interval contains the exact mixture sublevel set even when inversion stops early. The zero-count and all-count boundaries use their continuous likelihood values. Running intersections use endpoint maxima and minima and preserve the simultaneous-containment event; an empty intersection raises a violation and supplies no certificate.

Summary representation. A category-subset sum is enclosed both directly and by its simplex complement. The larger lower endpoint and smaller upper endpoint still enclose every law in the confidence rectangle intersected with the simplex. Monotonicity of q supplies the resolved-entropy bounds, additionally capped above by the resolved-mass upper bound times $\log 2$. Point category tables also use directed aggregate sums and an Arb entropy interval. Their "singleton" flag describes the categories, not exact real entropy. Approximate input normalization accepted by the population data layer is never taken as an exact simplex identity by the certifier.

Upper-search invariant and exits. Initially, every point of V_n is in the aggregate box. Directed geometry tests reject only impossible mass constraints, and the preceding information lower bound rejects only points outside the information budget. A split uses a representable midpoint inside its parent interval; its two closed children cover the parent, even when rounding makes one child identical to it. A retained box contributes the upward-rounded bound $\min(1, A_+ + u_+)$. Consequently the largest active or queued objective upper bound encloses the remaining supremum. Incumbents can raise this reported maximum but cannot lower it. Objective-gap, decision, box-width, stall and node-cap exits retain the relevant active and queued bounds. If arithmetic fails during a step, the active parent is retained even if only some children have been inserted. Failure before a conservative result is available propagates as a technical failure. When no boxes or incumbent remain, the reported upper bound is one. Thus convergence is unnecessary for domination of V_n.

Singleton shortcut. The scalar right-branch search is used only for point category inputs with point aggregate mass intervals. It additionally checks exact equality of those aggregates with the category sums and exact normalization with c. A failed identity check gives an assumption-free upper bound, without an incumbent. Exactly normalized tables with non-point aggregate sums use the box search; non-normalized point tables go directly to the conservative fallback. For fixed valid A, G, c, the largest relaxed endpoint uses exactly the supplied entropy upper endpoint T_U, not an independently recomputed nominal entropy. The initial right-branch lower bracket is at or above the exact minimizer $Ac/(A+G)$, and the upper bracket is c. Only a certified profile value above ρ can reduce the upper bracket. Convexity on this branch proves that every feasible hidden mass remains below it; an ambiguous sign cannot reduce that upper bracket. Upward conversion of $A + u_{\mathrm{hi}}$ gives the risk enclosure. Zero resolved or unresolved mass is covered by the continuous boundary formulas. The label `EXACT_SINGLETON` is used only when a verified lower incumbent equals the upper bound; otherwise the result is enclosed or a conservative fallback.

Feasible witnesses and gaps. Candidate allocation and the population root solver are proposal mechanisms only. For a proposed table, the certifier sums its supplied category entries exactly and defines terminal mass as the rational complement $c = 1 - A - G$. It checks exact category, aggregate, terminal and hidden-mass membership, including $0 \le u \le c$. The actual resolved entropy must lie inside its envelope, and an Arb upper bound on the actual table's information must be at most ρ. These checks construct a normalized original joint table regardless of the proposal's rounded aggregate metadata. Its risk is converted downward before becoming an incumbent. The singleton path applies the same entropy-membership and actual-information conditions after its exact mass identities. Failed or unresolved checks give no witness. Thus an incumbent

cannot exceed the original supremum and also belongs below the relaxed supremum. The reported gap is rounded upward from the enclosure minus this lower incumbent; no witness means no claimed gap. A lower incumbent above β refutes certification by this projection, without claiming that the true stream risk exceeds β.

Rejection searches and final assessment. The compatibility search assigns each box the downward bound $\max\{0, \inf q(A, G) - T_U\}$, which is below every original compatibility floor in that box. The intrinsic search uses a downward enclosure of $A_-/(A_- + G_+)$, taking zero if its denominator vanishes, and discards boxes only by the same safe geometry and information tests. This ratio bounds $A/(A + G)$ for every compatible original law retained in the box. In both searches, the minimum active or queued lower bound is therefore a global lower bound. Approximate point values can trigger stopping but never prune boxes or become the reported lower bound. At any early return the active box is included; after splitting its covering children are included. An empty retained queue returns zero rather than a point estimate. Exceptions propagate without a scientific result. If zero resolved mass remains plausible, the intrinsic bound is withheld. Finally, assessment checks that the projection and requested sensitivity budget agree. Rejections use strict comparisons with upward-rounded budget-plus-guard thresholds, while certification uses only $\widehat{U}_n \le \beta$ after the evidence gates. These facts compose the confidence input, summary enclosure, searches and assessment into the numerical condition of the article's time-uniform theorem under the stated assumptions.

The former population-midpoint shortcut and the archived floating wrappers are not covered retrospectively by this proof. Independent high-precision directional regressions check the repaired boundaries, entropy metadata, likelihood signs and search exits; passing those tests is supporting implementation evidence, not the enclosure argument itself. Historical experiment records are preserved without claiming that all their saved decisions were reverified under the revised code.

Compatibility rejection uses a lower bound on $q(A, G) - T_U$, rounded down and truncated at zero. Intrinsic rejection uses a lower bound on $A/(A + G)$ over compatible-plausible boxes. For example, on a box with $A_- + G_+ > 0$, $A_-/(A_- + G_+)$ is a valid lower bound. If zero resolved mass is plausible, intrinsic rejection is withheld. Strict comparison guards are 10^{-12}. Population root tolerance is 10^{-12}; identity residual tolerance is 10^{-10}; categorical inversion tolerance is 10^{-12}; the nominal outer objective gap is 10^{-6} with a 2,000,000-node cap. A normalized box-width exit or stall is reported separately from objective convergence.

5 Complete confirmatory outcomes and design

All rows in Table S1 have 100 independent streams, $\delta = 0.05$, and confirmatory indices 1,000,000–1,000,099. Within a law, the same finest-resolution streams feed each partition. The ordinary grid is 2,000, 5,000, 10,000, 20,000; the power and matched-control grids and low-prevalence grid are given in the article. Dense monitoring uses 250, 500, 750, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 3,500, 4,000, 4,500, 5,000, 5,500, 6,000, 6,500, 7,000, 7,500, 8,000, 8,500, 9,000, 9,500, 10,000. Endpoint-uncertifiable uses 10,000, 20,000, 50,000. Zero counts include all streams and do not imply zero certification probability.

Table S1: All 32 final-cell certification outcomes. Intervals are conditionwise two-sided exact 95% intervals for ever certifying by the horizon, using 100 trials per row.

Scenario	K	ρ	β	Horizon	Certified	95% interval
Endpoint-uncertifiable	1	0.0800	0.070	50,000	0/100	[0.000, 0.036]
Endpoint-uncertifiable	2	0.0800	0.070	50,000	100/100	[0.964, 1.000]
Endpoint-uncertifiable	4	0.0800	0.070	50,000	100/100	[0.964, 1.000]
Endpoint-uncertifiable	8	0.0800	0.070	50,000	99/100	[0.946, 1.000]
Population decision boundary	1	0.0500	0.060	20,000	0/100	[0.000, 0.036]
Population decision boundary	8	0.0500	0.060	20,000	0/100	[0.000, 0.036]
Ignorable delay	1	0.0200	0.090	20,000	98/100	[0.930, 0.998]
Ignorable delay	8	0.0200	0.090	20,000	0/100	[0.000, 0.036]
Intrinsic impossibility	1	0.0500	0.050	20,000	0/100	[0.000, 0.036]
Low prevalence	1	0.0200	0.050	20,000	100/100	[0.964, 1.000]
Low prevalence	2	0.0200	0.050	20,000	99/100	[0.946, 1.000]
Low prevalence	4	0.0200	0.050	20,000	99/100	[0.946, 1.000]
Low prevalence	8	0.0200	0.050	20,000	99/100	[0.946, 1.000]
No timing control	1	0.0800	0.090	50,000	94/100	[0.874, 0.978]
No timing control	2	0.0800	0.090	50,000	28/100	[0.195, 0.379]
No timing control	4	0.0800	0.090	50,000	0/100	[0.000, 0.036]
No timing control	8	0.0800	0.090	50,000	0/100	[0.000, 0.036]
Misspecified sensitivity	1	0.0200	0.045	20,000	0/100	[0.000, 0.036]
Misspecified sensitivity	2	0.0200	0.045	20,000	0/100	[0.000, 0.036]
Incompatible budget	4	0.0200	0.060	20,000	16/100	[0.094, 0.247]
Incompatible budget	8	0.0200	0.060	20,000	0/100	[0.000, 0.036]
Near degeneracy	1	0.0493	0.050	20,000	99/100	[0.946, 1.000]
Near degeneracy	8	0.0493	0.050	20,000	99/100	[0.946, 1.000]
Dense monitoring	1	0.0200	0.065	10,000	0/100	[0.000, 0.036]
Strong timing	1	0.0800	0.090	50,000	96/100	[0.901, 0.989]
Strong timing	2	0.0800	0.090	50,000	100/100	[0.964, 1.000]
Strong timing	4	0.0800	0.090	50,000	100/100	[0.964, 1.000]
Strong timing	8	0.0800	0.090	50,000	99/100	[0.946, 1.000]
Risk near budget	1	0.0500	0.052	20,000	0/100	[0.000, 0.036]
Risk near budget	8	0.0500	0.052	20,000	0/100	[0.000, 0.036]
Unsafe law	1	0.0500	0.040	20,000	0/100	[0.000, 0.036]
Unsafe law	8	0.0500	0.040	20,000	0/100	[0.000, 0.036]

Table S2: Current search states among 96 decision-targeted streams at selected checkpoints. Four full-search representatives per cell are excluded from these rows.

Scenario	K	Events	Certified	Refuted	Unresolved	Stopped
Strong timing, $\beta = .09$	1	5,000	0	89	7	0
Strong timing, $\beta = .09$	2	5,000	96	0	0	0
Strong timing, $\beta = .09$	1	50,000	92	0	4	0
Matched control	2	50,000	27	1	67	1
Matched control	8	50,000	0	94	0	2

Each row sums to 96. These are states at the specified checkpoint, whereas the headline proportions count all streams that have ever certified among 100, including the four full-search representatives. Decision-targeted execution can stop when a bound settles the risk-budget comparison; full search continues toward its objective-gap target. A full search can report a stall after already obtaining a certifying upper bound. Here, refutation denotes the recorded decision-refuted termination, conditional on a numerically verified feasible lower incumbent above β; it withholds certification and does not establish that the true risk exceeds β. The complete feasibility contract is not established for the archived implementation. Unresolved denotes a stalled

search that neither certified nor supplied a refutation; further search need not certify. Stopped paths have no current update after their running confidence region became empty, but remain in all denominators. No model-incompatible state was recorded at any of the 160 power checkpoints.

Additional laws use the parameter order $(\theta, q_1, q_0, \lambda_1, \lambda_0)$. High unresolvedness has $(.05, .70, .40, .35, -.10)$. The timing-only law has $(.05, .10, .10, .45, -.15)$ and the terminal-only law has $(.05, .30, .05, 0, 0)$. Near degeneracy has $(.01, .90, .01, .80, -.80)$; intrinsic impossibility has $(.15, .10, .10, 0, 0)$. The unsafe, risk-near-budget, incompatible-budget, misspecified-sensitivity and population-decision-boundary scenarios use the ordinary timing-and-terminal harmful-late law from the article. The ignorable-delay and dense-monitoring scenarios use the independent-resolution law. The 32-band validity law is regenerated at 32 bands with the ordinary timing-and-terminal parameters, rather than being a coarsening of the eight-band law.

In the seven validity scenarios, $\rho = I_{\text{true}} + 0.01$, except the minimum-information completion, which sets terminal harmful mass to Ac/r and uses $\rho = \tau + 0.002$. The near-certification risk budget is its own sharp upper bound at the designated ρ plus 0.002; other validity budgets are 0.05. Each scenario uses 500 independent streams, seed indices 0–499, through 500 matured events. The nominal error budget is unchanged. Construction exclusions are monitored at every event and risk-bound diagnostics at the five 100-event checkpoints.

A separate descriptive assessment of the longer-horizon power records retains the longest observed horizon for each of 21 distinct law–partition stream sets. Across 2,100 indexed stream outcomes, 58 exclusions of the true categorical law were recorded (2.8%), with at most six exclusions among 100 streams in any set. These observations are not 2,100 independent Bernoulli trials: some partitions share event streams, and empty confidence regions terminate further updates. Accordingly, this aggregate is descriptive and is not assigned a pooled binomial confidence interval or a campaign-wide coverage interpretation.

For a count x of m streams, the two-sided 95% binomial interval has lower endpoint zero if $x = 0$ and otherwise $B^{-1}_{x,m-x+1}(0.025)$, and upper endpoint one if $x = m$ and otherwise $B^{-1}_{x+1,m-x}(0.975)$. One-sided violation limits use $B^{-1}_{x+1,m-x}(0.95)$. This distinguishes the Monte Carlo intervals from the stream's own time-uniform confidence region. The prespecified final-horizon power target uses the two-sided interval's lower endpoint, exactly as stored in the publication artifacts.

Repeated-static monitoring uses per-category Wilson intervals with $z = \Phi^{-1}\{1-\delta/(2d)\}$ and no correction across looks. It can therefore have greater power without sharing TrajCert's exact time-uniform guarantee. At the final horizon of the valid ignorable-delay regime, its counts are 98/100 at $K = 1$ and 0/100 at $K = 8$, while the ignorable-delay reference certifies 98/100 at either partition. In the unsafe law, the latter reference certifies falsely in 7/100 at each displayed partition, but its independence assumption fails there. These are assumption-violation diagnostics. In the dense-monitoring scenario it certifies 97/100, whereas neither sensitivity projection certifies; the stronger identifying assumption changes the population target bound.

All TrajCert power cells record zero false risk certification. The stored premature-certification counter is also zero: it counts certificates in cells classified as compatible but not population-certifiable, and does not include model-incompatible cells. The sensitivity-misspecified two-band cell is population-certifiable under its assumed model but has true risk $0.05 > 0.045$ and true partition information above $\rho = 0.02$; its observed zero certifications do not validate that assumption. The incompatible four-band cell certifies 16/100 although $\rho < \tau$ in the population, with true risk below its budget. These outcomes illustrate the distinction between risk error and a certificate issued under a false structural assumption.

6 Reproducibility record

The canonical configuration is `configs/trajcert.yaml`; its SHA-256 is

`865eeb27e1bdbead6198ecfc1f45349a483fe44e4ea6a06e71012ca0e35058b7.`

It is byte-identical to the locally retained 19 October 2026 frozen configuration copy. This comparison establishes present byte equality, not independent historical attestation of the source code used by every experimental run. The scientific specification digest is

`3e9e0c6584a3b4f84bcec4575bf9392de841fe3a47eb9bd7f33e86f7eeb81721.`

The report records commit `36e76519cc6f2496c8b6d0a93e7abae2cf4303af`; the manuscript inspected the clean checkout at `6a37400fc10e977d4c568d7e94d88f3f49dd67aa`. These identifiers serve different roles.

The dependency fingerprint does not hash the source tree, so the report commit is not proof that every experiment used an identical implementation. Power precedes a later runtime optimization, whereas coverage uses the optimized code. The campaign reports identical serialized scientific-output digests, excluding runtime fields, on 11 selected cases; a complete per-cell equivalence rerun is not available. No experimental rerun was performed during manuscript preparation.

The environment lock digest recorded in the report is

`2f94341d20401569cbbd305333d78becd88a046ca1b272c30b49997836fcf9db.`

The software manifest specifies Python 3.12 or later, NumPy 2.5.2, SciPy 1.17.1, mpmath 1.3.0 and python-flint 0.9.0, with the canonical resolved graph in `uv.lock` and hash-locked export in `requirements.lock`. These are declared dependencies; the archived timing data do not independently identify the physical CPU or a reproducible machine load.

Seeds derive from SHA-256 of a semantic event-stream namespace and stream index, using NumPy's PCG64 generator. The namespace includes the law and its finest band count, and excludes the analysis partition, sensitivity budget, risk budget and experiment name. Coarse analyses therefore reuse the same stream; shorter horizons use prefixes. Law selection used population search and separate exploratory streams, and the confirmatory configuration was frozen before the reported streams. This separation prevents tuning on the confirmatory streams, while leaving the designed-law scope of the experiment explicit.

The exploratory design included ten-stream pilots, with twenty streams used for the optional-stopping grid. A separate ten-stream pilot of the ordinary population-decision example reported no TrajCert certification through 200,000 matured events. This pilot is distinct from the confirmatory evaluation and is not included in its certification estimates. The exploratory observation is algorithm-dependent and does not establish that population certification is unattainable at a larger finite-sample horizon. The locally retained pilot records are private and are not included in the distributed evidence archive.

The manuscript-side checks read existing scientific JSON, Parquet and CSV files. They verify the 116 source/destination pairs in the report manifest, all 320 power batches, original stream indices, every checkpoint certification count, the exact binomial intervals, all seven coverage completion markers and batch denominators, the frozen configuration, scaling-unit conversions and 176 targeted/full decisions. Main figures are rendered solely from these existing numerical sources. The archived final test log records 961 passing tests; this is a historical validation record, not a newly executed test suite. The accompanying evidence archive contains unchanged scientific-result JSON files, individual power-batch outcomes, validity-batch records and exported CSV tables for these three current studies, with a SHA-256 manifest and a link to the inspected public code commit. Internal manuscript audits, historical experiments and downloaded literature are excluded from the submission package.